Team LeafScan

Real-Time Leaf Detection & Species Classification System User Manual

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Introduction

This manual describes the features available to users of LeafScan, a web platform that finds a single leaf in a photo and identifies its species, and gives a detailed explanation of how to create an account, upload a leaf image, read the result and review past scans. Administrators will also find the workflows for managing users and reviewing platform activity.

LeafScan is built from four parts: a React web interface, a Node.js/Express API (login, roles, uploads), a Python/FastAPI AI service (YOLOv8 leaf detection and a MobileNetV2 species classifier) and a MongoDB database.

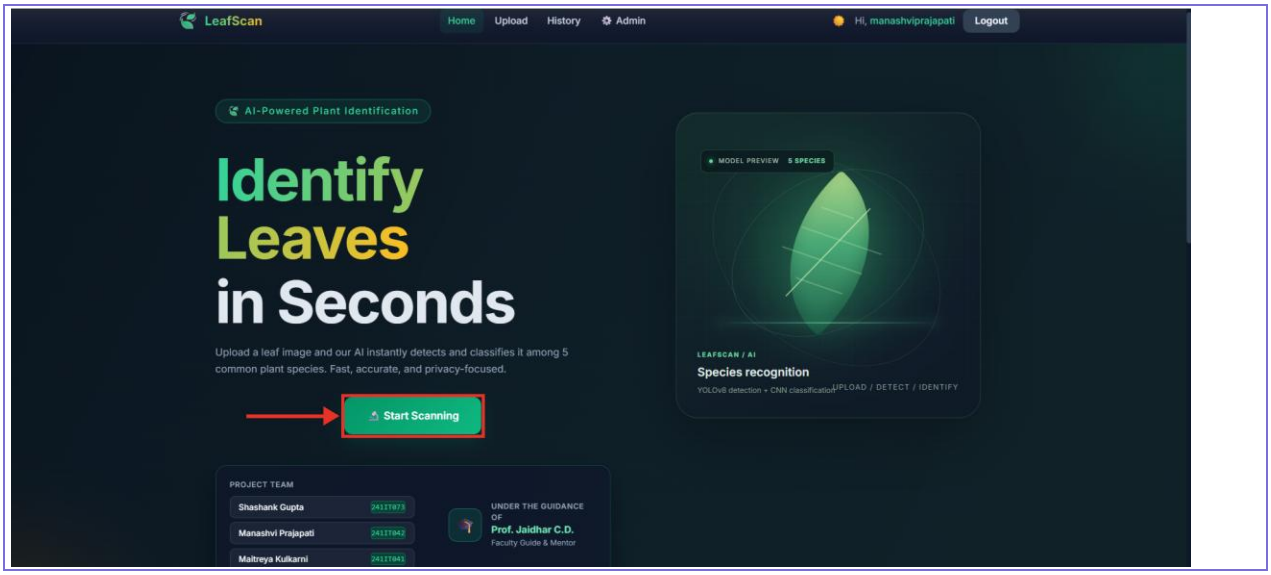
Supported species

The system is trained to identify five species.

Common name	Scientific name	Leaf characteristics
Guava	Psidium guajava	Elliptic/oval blade, prominent parallel veins, dark green
Lemon	Citrus limon	Elliptic to ovate, subtle crenulate margins, citrus aroma
Lychee	Litchi chinensis	Glossy pinnate leaflets, lanceolate blade, smooth margin
Mango	Mangifera indica	Long lanceolate leaves, wavy margins, deep green gloss
Olive	Olea europaea	Narrow lanceolate shape, silvery-grey underside

Accessing the system

1. Open a modern web browser (Chrome 100+, Firefox 100+, Edge 100+ or Safari 15+).
2. Enter `http://localhost:5173` in the address bar.
3. The LeafScan landing page appears with the navigation bar and system highlights.



In a standard deployment the platform components run at these addresses:

Component	Address
Web application	`http://localhost:5173`
REST API server	`http://localhost:5000/api`
AI service API docs	`http://localhost:8000/docs`

Actions available to the user

A registered user has access to the following actions within LeafScan:

- Create an account, verify the email address and sign in.
- Upload a single leaf image and identify its species.
- View the species, confidence score and botanical summary of each scan.
- Browse personal scan history (thumbnails, species, confidence, timestamp) page by page.
- Download personal scans as a CSV report.
- Reset a forgotten password by email.
- Switch between dark and light themes.

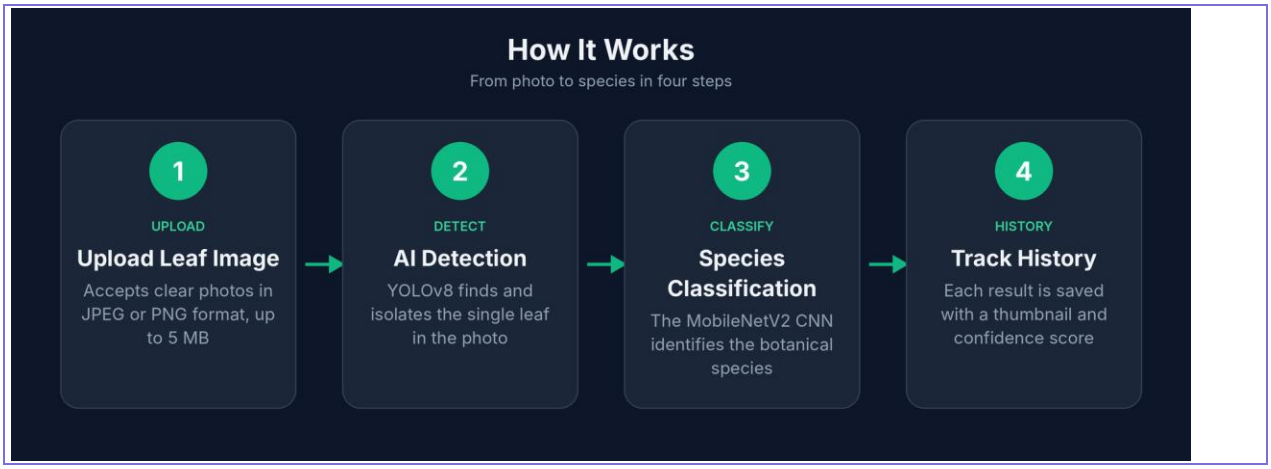
Access depends on the role of the account:

Feature	Guest	Registered user	Administrator
Browse landing page and species info	Yes	Yes	Yes
Register, verify, sign in, reset password	Yes	Yes	Yes
Upload and scan leaf images	No	Yes	Yes
View predictions and confidence	No	Yes	Yes
View own history and export CSV	No	Yes	Yes
Admin dashboard (`/admin`)	No	No	Yes
Platform statistics, user accounts, audit logs	No	No	Yes
View other users' leaf images or thumbnails	No	No	No

Other users' leaf images and thumbnails are never visible to anyone, including administrators (privacy preserved).

LeafScan module – Overview

The landing page explains the four stages every scan goes through:



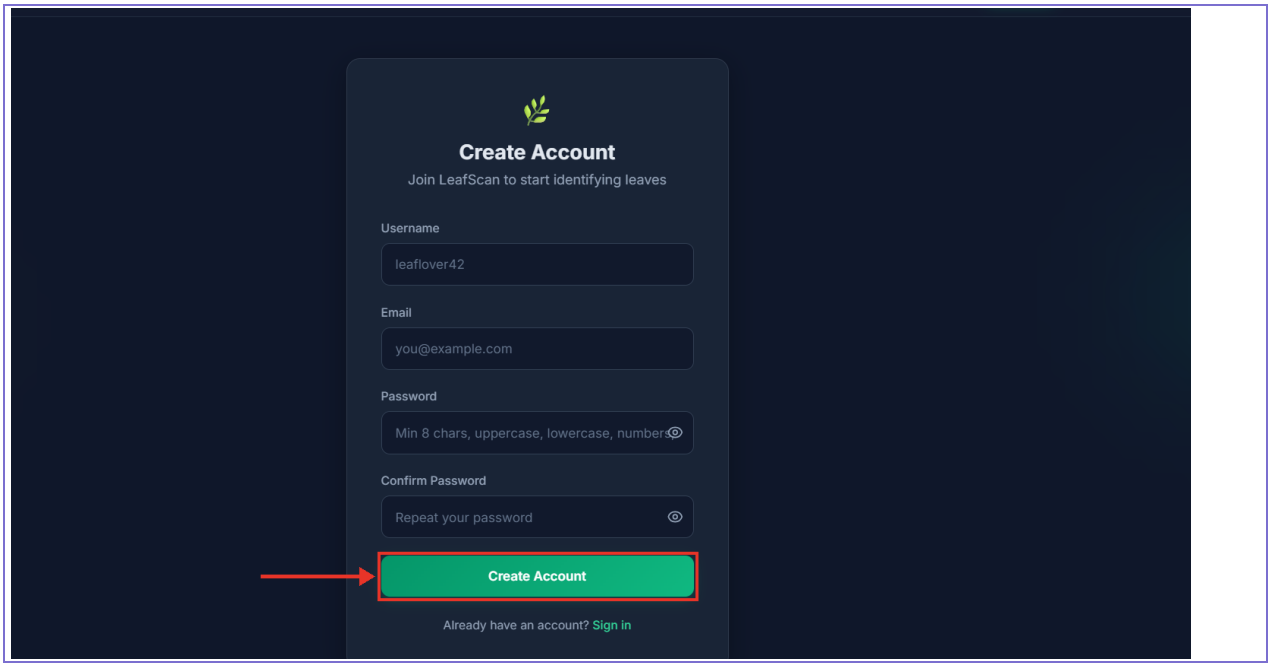
Scroll to the **Supported Species** section of the landing page to see cards for Guava, Lemon, Lychee, Mango and Olive.

LeafScan module features

1. Creating an account

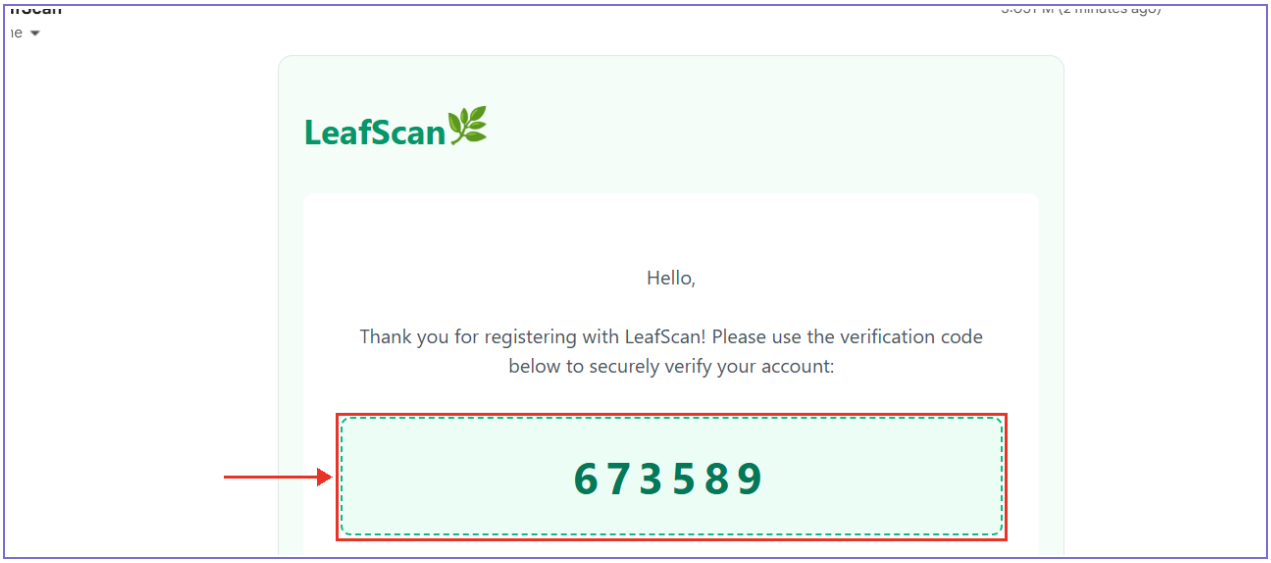
1. Click **Sign Up** in the top-right corner of the navigation bar, or **Get Started Free** on the landing page.

2. Fill in the registration form:
    - **Username:** unique, 3–30 alphanumeric characters.
    - **Email:** a valid address; it receives the verification code and recovery messages.
    - **Password:** at least 8 characters, with uppercase, lowercase, a number and a special symbol.
    - **Confirm Password:** must match exactly.
3. Watch the **Password Strength Indicator**, which rates the password Weak, Good or Strong.
4. Click **Create Account**.



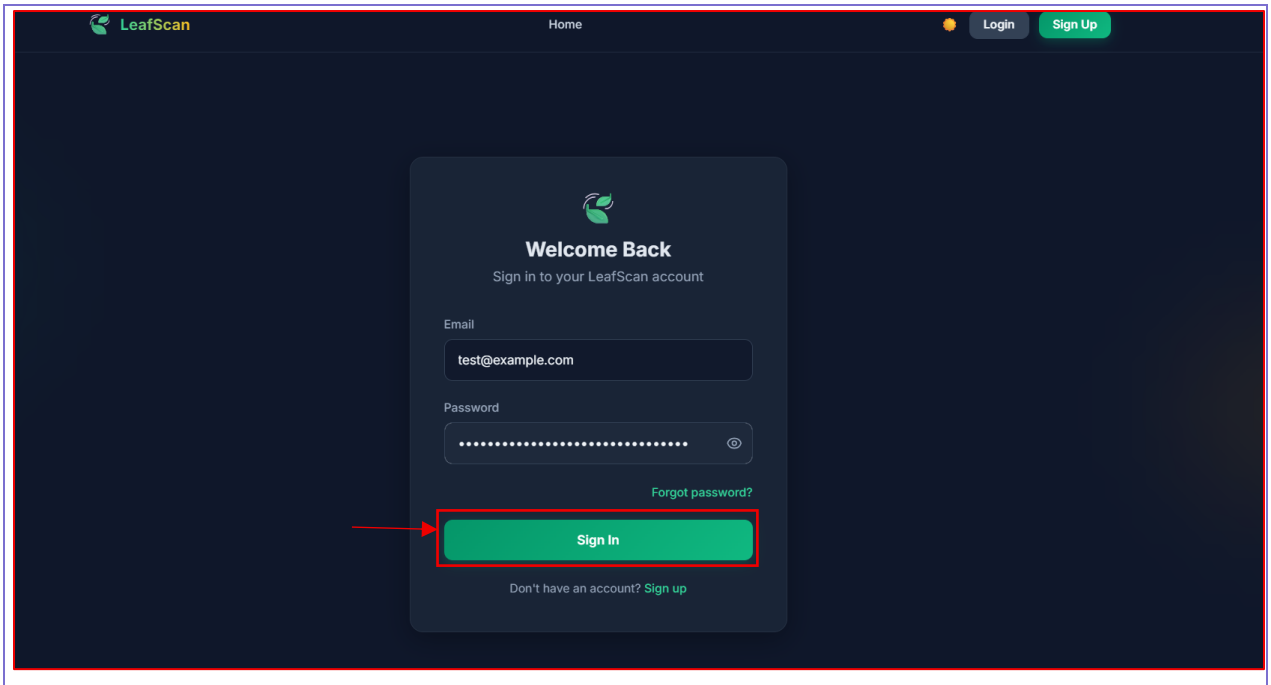
2. Verifying your email (OTP)

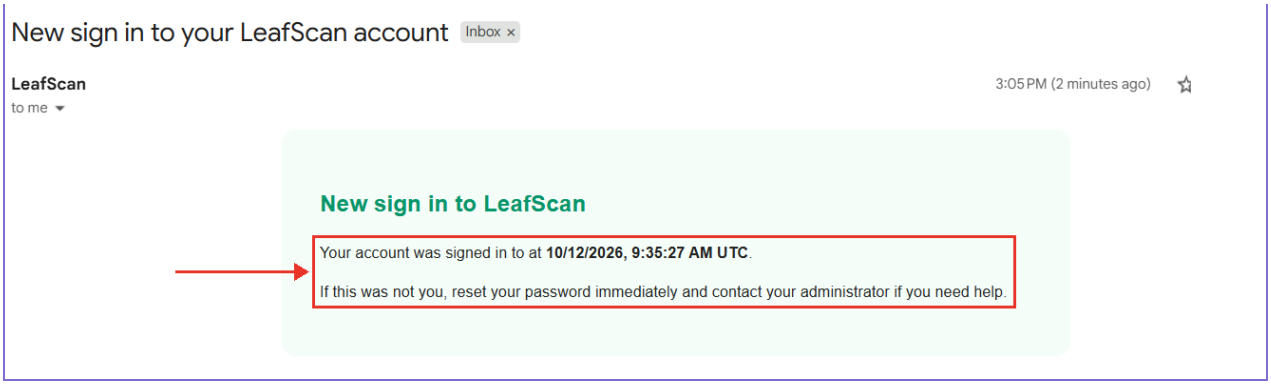
1. After you submit the form, the **Verify Your Email** page opens.
2. A 6-digit code is sent to your email address in a message from LeafScan (shown below).
3. Type the code into the **Verification Code** box and click **Verify**.
4. If the code does not arrive, click **Resend** to get a new one.
5. On success your account becomes active and you are taken to the Upload page.



3. Signing in

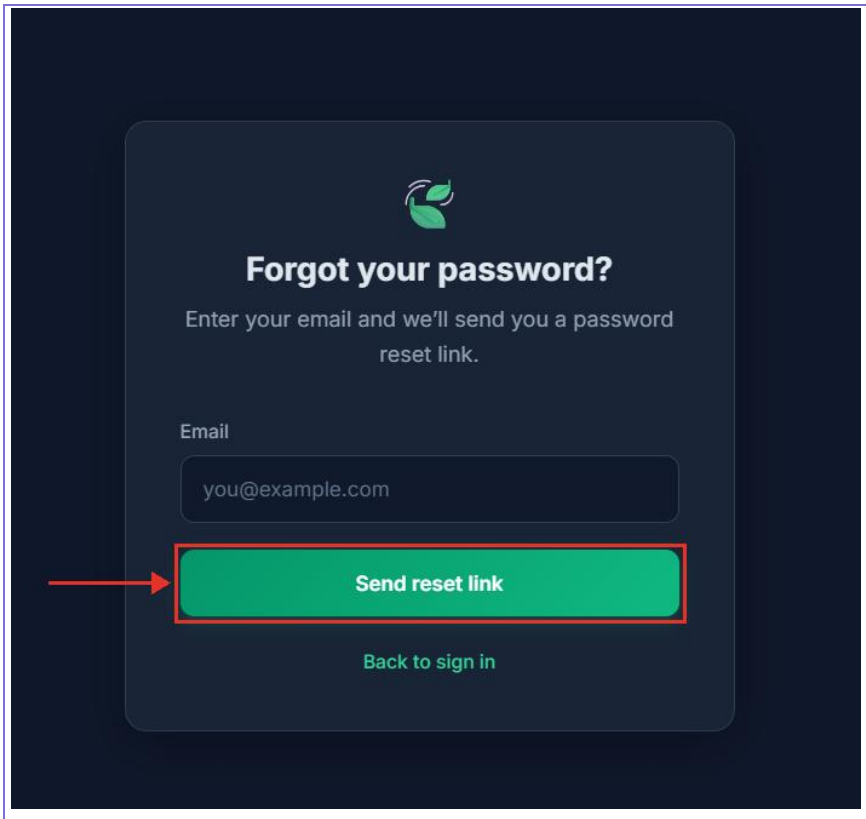
1. Click **Login** in the navigation bar.
2. Enter your registered email and password. Use the eye icon to show or hide the password.
3. Click **Sign In**.
4. The navigation bar now greets you (“Hi, username”) and shows the **Upload** and **History** tabs.
5. Each sign-in also sends a “New sign in to your LeafScan account” email with the time. If it was not you, reset your password immediately and contact your administrator.

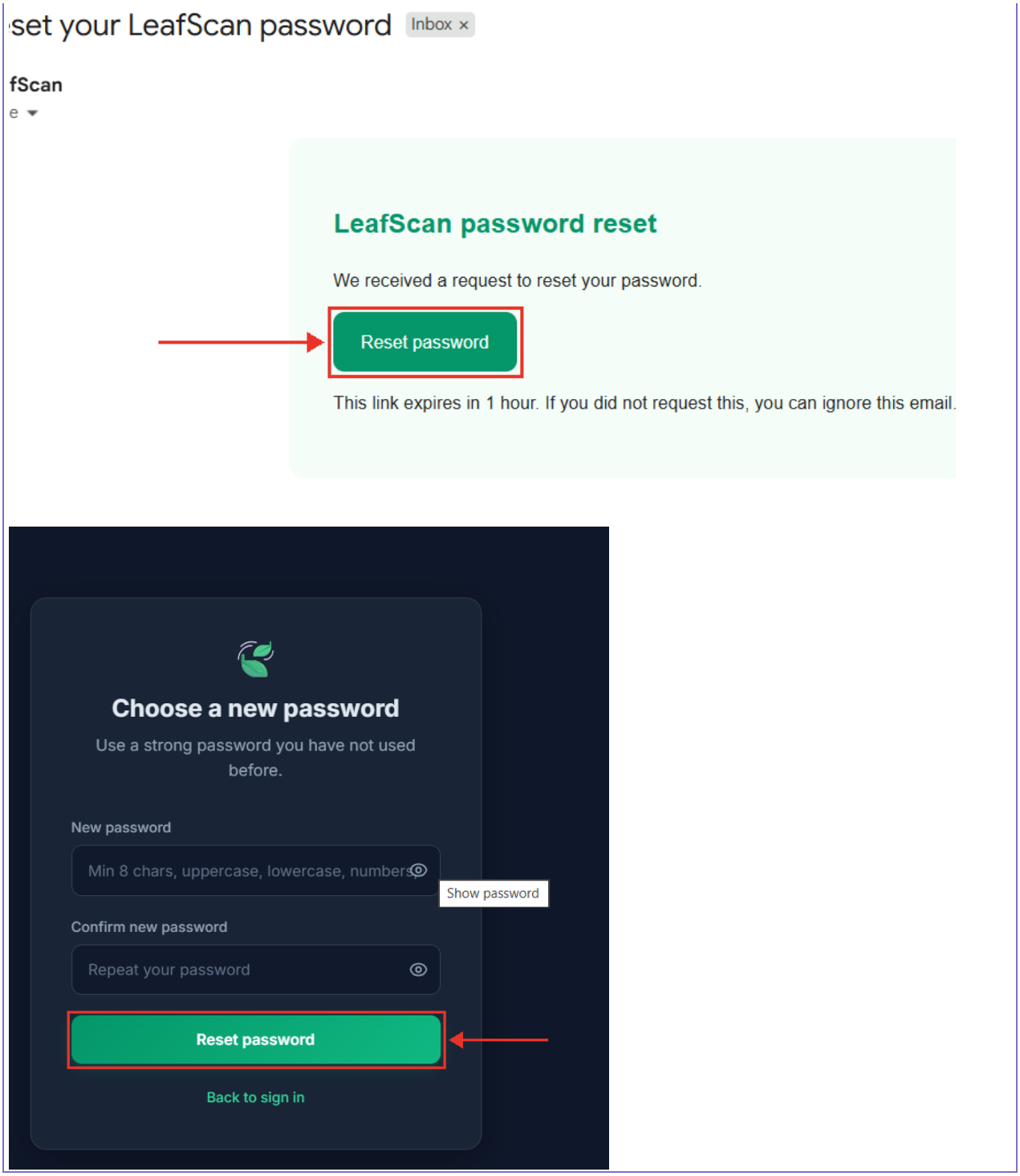




4. Resetting a forgotten password

1. On the Login screen, click **Forgot password?**.
2. Enter the email linked to your account and click **Send reset link**.
3. Open the email and click **Reset password** (the link expires in 1 hour) to reach the **Choose a New Password** page.
4. Enter the new password twice and click **Reset password**.
5. Sign in with the new password once the confirmation appears.



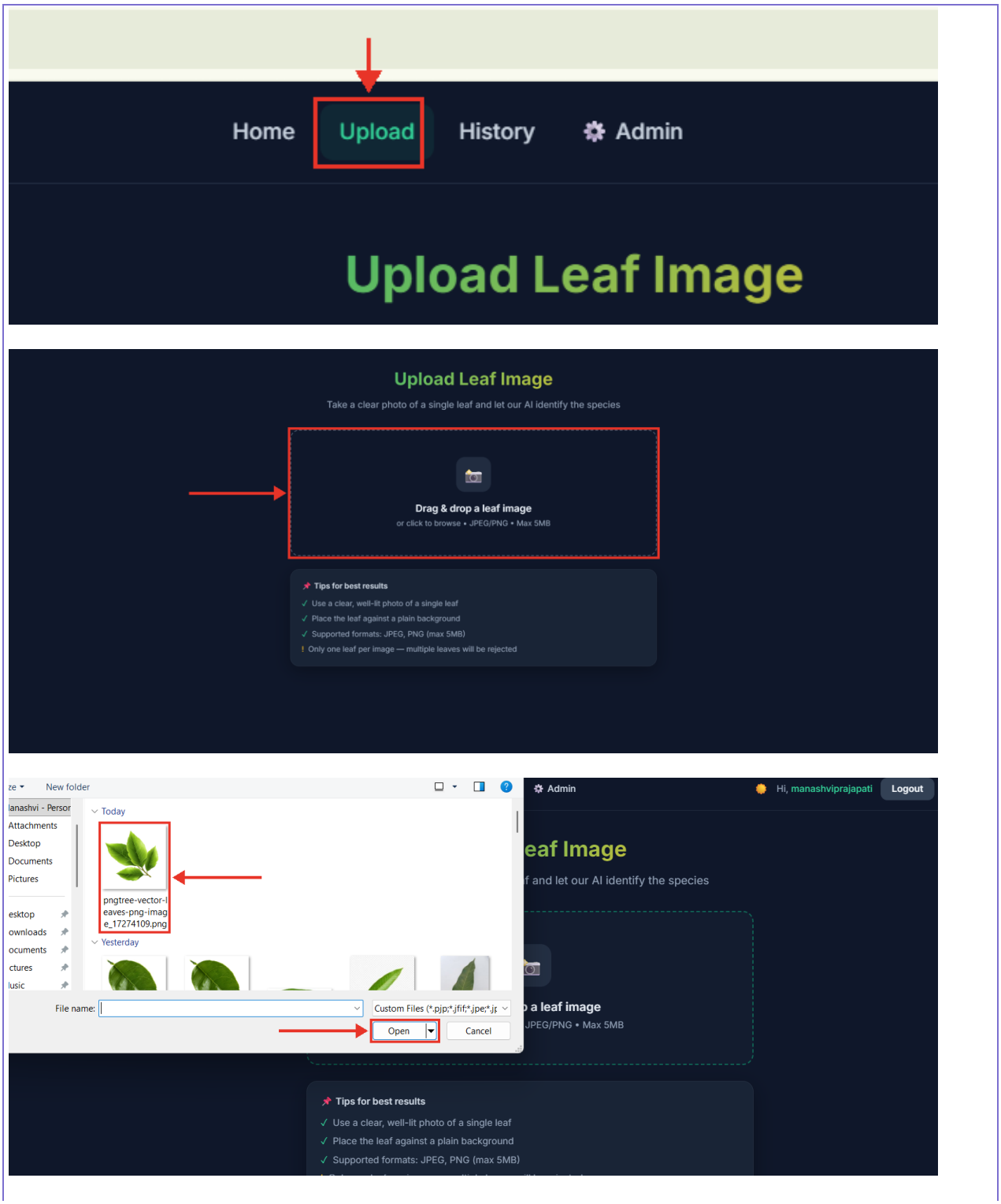


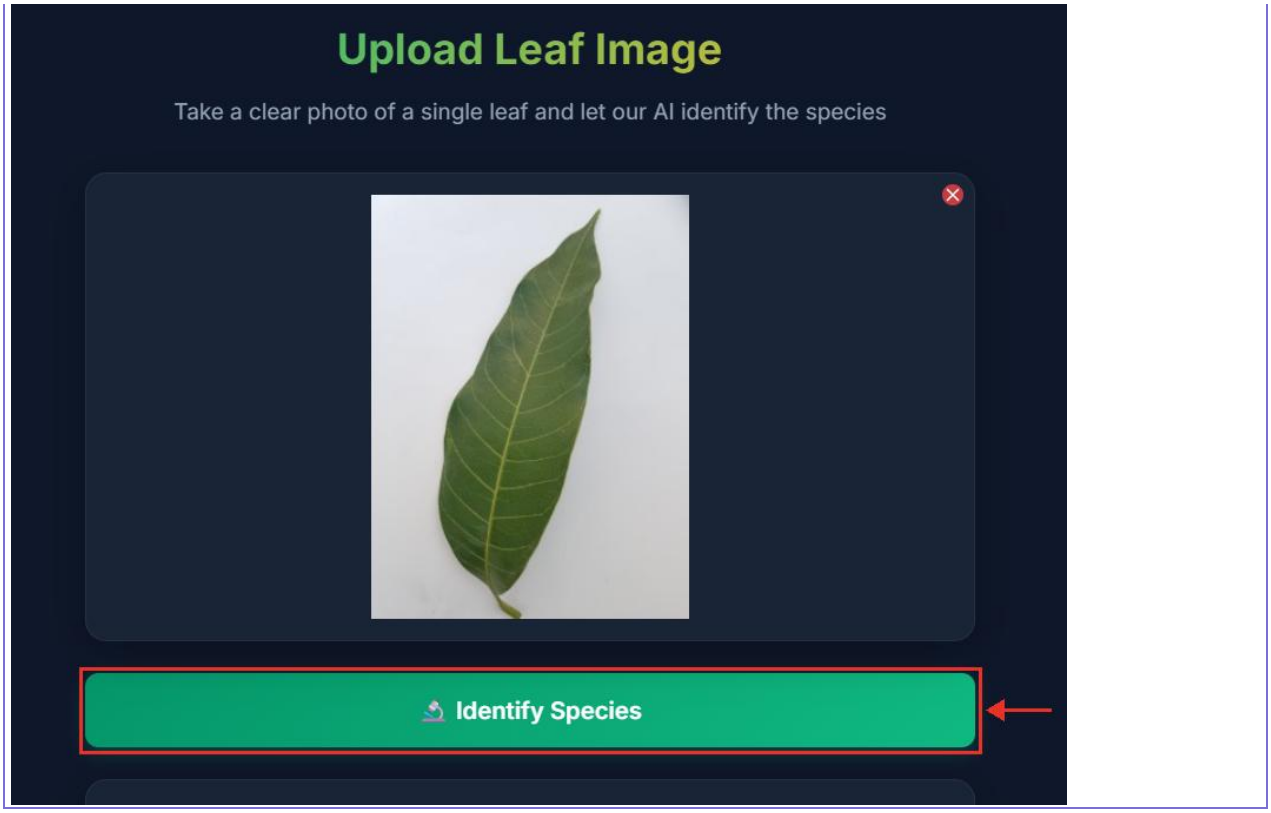
5. Uploading a leaf image

1. Click **Upload** in the navigation bar.
2. Add your image in one of two ways:
    - **Drag and drop** a file onto the dashed box.

- **Click** inside the box to open the file browser.

3. A large preview appears, with a red ✕ button to remove the image and choose another.
4. Click **Identify Species** to start processing.





6. Photo requirements

Follow these rules for accurate results:

- **Leaf count:** exactly one leaf. Photos with no leaf or several leaves are rejected.
- **Background:** a plain, contrasting surface such as white paper or a table.
- **Format and size:** JPEG (`.jpg`, `.jpeg`) or PNG (`.png`), at most 5 MB.

7. How your photo is processed

Results normally appear in under 3 seconds. Behind the scenes:

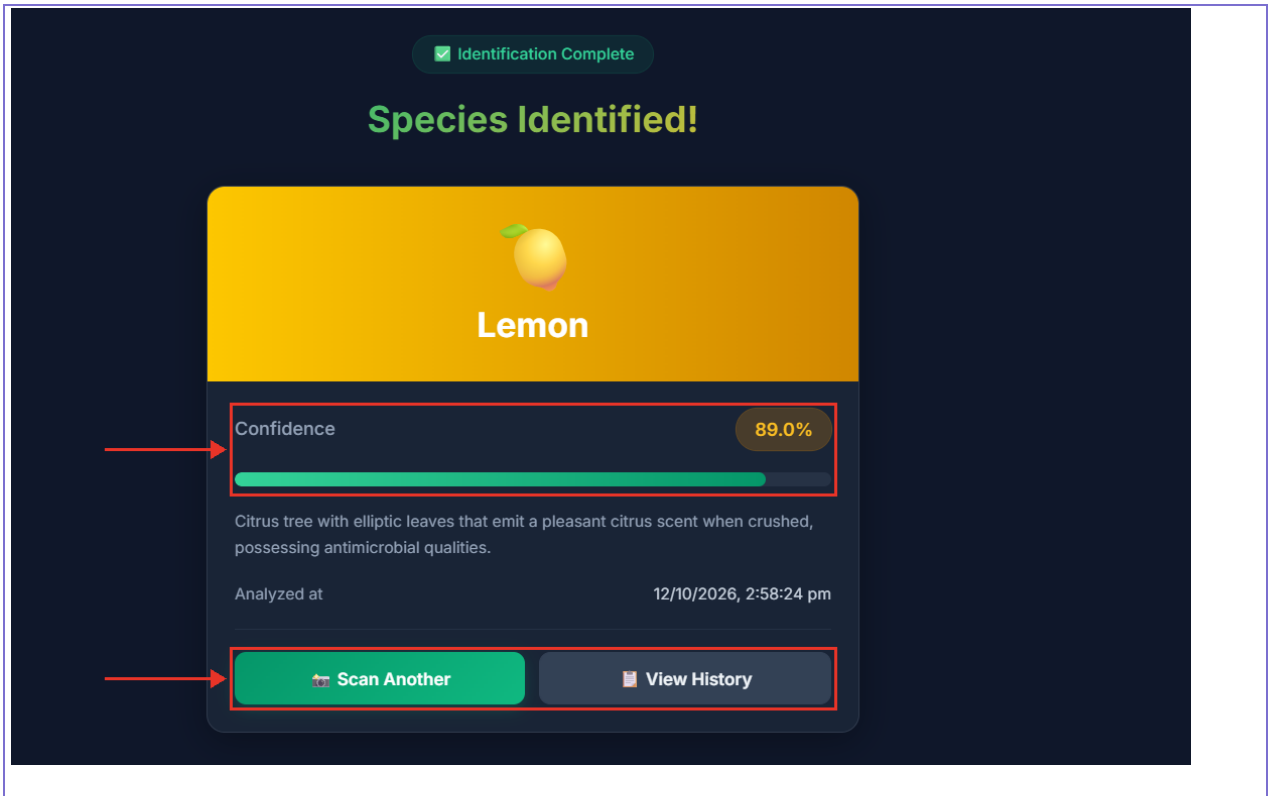
1. The image is sent to the server.
2. YOLOv8 counts the leaves; anything other than exactly one leaf is rejected.
3. The leaf is cropped to its bounding box.
4. The MobileNetV2 classifier identifies the species.
5. A 150×150 thumbnail and the result are saved to your history.
6. The full-resolution photo is discarded from memory immediately.

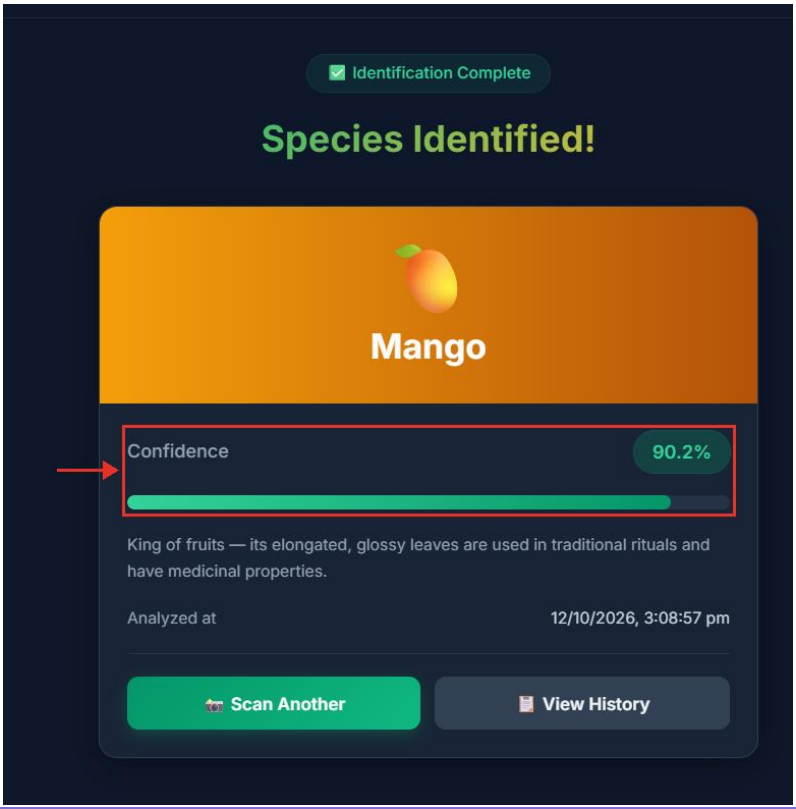
8. Reading the result

The **Result** page shows:

- **Species name and emoji:** Guava, Lemon, Lychee, Mango or Olive.
- **Confidence badge and meter** with the exact percentage.
- **Botanical summary** covering leaf shape, veins and traditional uses.
- **Analysis timestamp** in local time.
- **Scan Another** (back to Upload) and **View History** buttons.

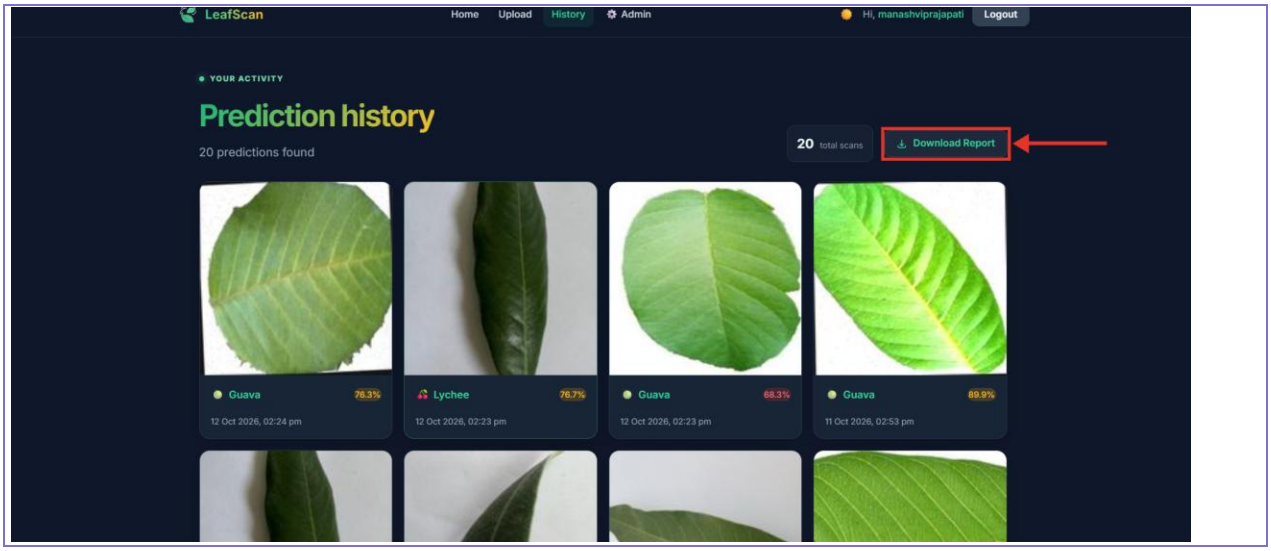
Confidence	Colour	Meaning
90% or higher	Green	High confidence
70% to 89%	Amber	Moderate confidence
Below 70%	Red	Low confidence; retake the photo





9. Viewing scan history

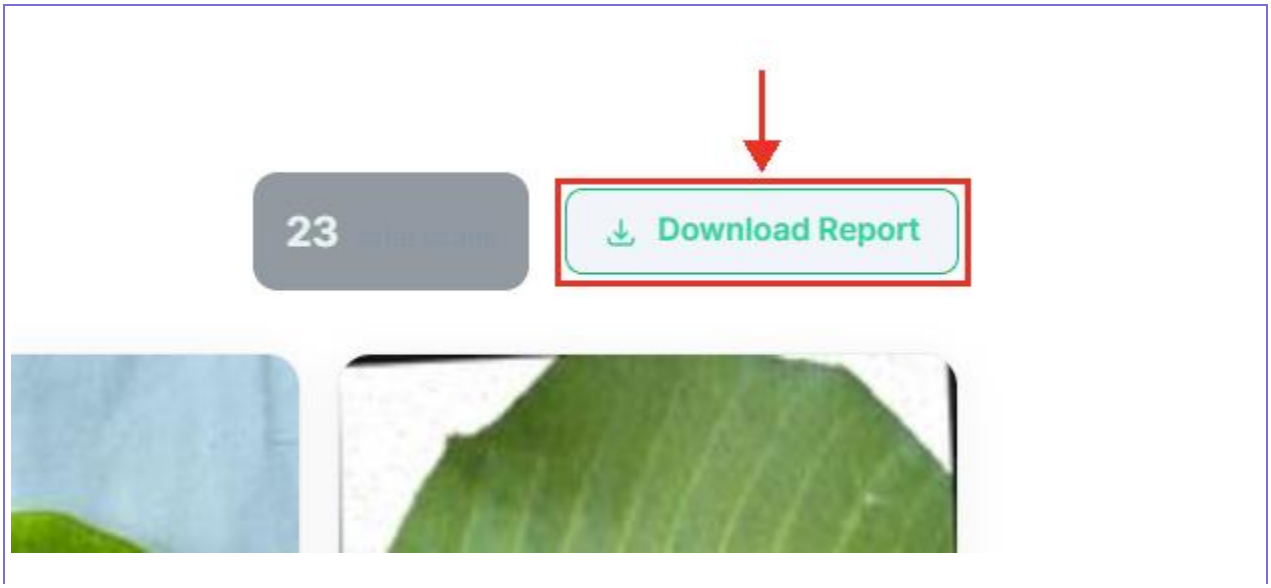
1. Click **History** in the navigation bar.
2. Check the **Total Scans** counter at the top.
3. Browse the cards, each showing the thumbnail, species, confidence badge and timestamp.
4. Use the page numbers and arrows at the bottom to reach older scans.



10. Downloading your activity report

1. On the **History** page, click **Download Report** next to the scan total.
2. A file named `leafscan_report.csv` is saved to your computer.
3. It contains four columns:

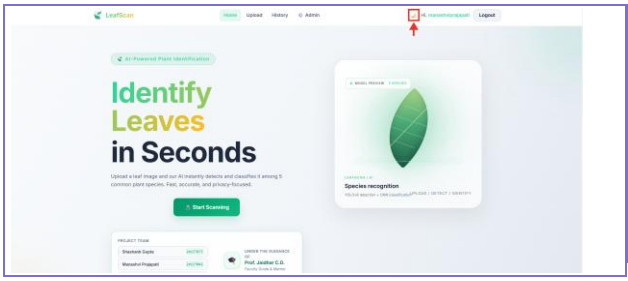
Column	Content
Prediction ID	Unique database identifier
Species	Identified plant species
Confidence Score (%)	Model confidence
Timestamp	Date and time of the scan



	A	B	C	D	E	F
	12/10/202	Mango	39.39			
	12/10/202	Lemon	88.95			
	12/10/202	Guava	76.31			
	12/10/202	Lychee	76.73			
	12/10/202	Guava	68.32			
	11/10/202	Guava	89.93			
	11/10/202	Lychee	68.65			
0	11/10/202	Olive	80.44			
1	11/10/202	Mango	90.17			
2	11/10/202	Guava	66.59			
3	11/10/202	Guava	27.81			
4	11/10/202	Lemon	84.83			
5	11/10/202	Guava	48.27			

11. Switching theme

Click the sun/moon button on the right of the navigation bar to switch between the dark theme (slate and emerald) and the light theme (white and emerald). Your choice is remembered in the browser between visits.



Light theme



Dark theme

NOTE:

- Full-resolution photos are never stored. Only a 150×150 thumbnail is kept, and only you can open it.
- You can submit at most 10 prediction requests per minute.
- Administrators see metadata only: username, species, confidence and timestamp.

Error messages and troubleshooting

Message	Cause	What to do
No leaf detected in the image.	No recognizable leaf, or lighting too dim	Place a fresh leaf flat on a clean, light surface and retake the photo
Multiple leaves detected (N).	More than one leaf in the frame	Keep only one leaf visible, or crop out the others
File too large. Maximum size is 5MB.	Image over 5 MB	Compress or resize it in a photo editor
Invalid file type. Only JPEG and PNG...	Unsupported format such as `.webp` or `.pdf`	Save the image as `.jpg` or `.png`
Too many prediction requests...	More than 10 requests in one minute	Wait 60 seconds, then try again
Leaf detection service is temporarily unavailable.	AI service on port 8000 is offline or starting	Ask the administrator to start it (`uvicorn main:app --port 8000`)

Frequently asked questions

Why are my full-resolution photos not in History?
LeafScan processes uploads in temporary memory and deletes them right after identification. Only the compressed thumbnail is saved.

Can administrators see my leaf images?
No. Thumbnails can be opened only with the owner's credentials. Administrators see classification metadata only.

Why is my confidence score low?
The leaf may be wilting, damaged or affected by glare, or it may belong to a species outside the five supported ones.

How do I change my password while signed in?
Use **Forgot password?** on the sign-in screen to get a reset link by email.

Revision history

Version	Date	Summary	Authors
1.0	12/10/2026	Initial release covering sign-up, YOLOv8 detection, CNN classification, history export and administration	Shashank Gupta, Manashvi Prajapati, Maitreya Kulkarni

Project guide: Prof. Jaidhar C.D. · Department of Information Technology, National Institute of Technology Karnataka (NITK), Surathkal